PARENT ONLY — SOLVED ANSWER KEY

Vyom · Parent answer key

18 Jan – 24 Jan 2027 · Monday–Sunday · NDPS Class 9

Vyom ko test se pehle ye PDF mat dena. Student pack mein sirf questions aur blank answer space hai. Is alag parent PDF mein har original question ke saath uska matching solved answer, working aur marking points diye hain; Test ID se student paper match karein.

Checking: same evening. C = concept, R = recall, K = careless, T = time.

90%+: short revision. 75–89%: weak-topic practice + retest. Below 75%: topic dobara samjhao; mistakes notebook mein likho.

Answer keys existing source question banks se li gayi hain; school pattern abhi unconfirmed hai. Alternate valid answers ko marking guidance ke mutabik accept karein.

Week score register

Date / test ID	Test	Marks	Galti / retest
18 Jan D2027-01-18	Daily mixed Daily test	___ / 10	Retest: 21 Jan
19 Jan D2027-01-19	Daily mixed Daily test	___ / 10	Retest: 22 Jan
20 Jan D2027-01-20	Daily mixed Daily test	___ / 10	Retest: 23 Jan
21 Jan D2027-01-21	Daily mixed Daily test	___ / 10	Retest: 24 Jan
22 Jan D2027-01-22	Daily mixed Daily test	___ / 10	Retest: 25 Jan
23 Jan D2027-01-23	Daily mixed Daily test	___ / 10	Retest: 26 Jan
24 Jan T2027-01-24	Maths Full paper (3 hr)	___ / 80	Retest: 27 Jan

Daily Test · Answer key

Monday, 18 January 2027 · 10 marks

Test ID: D2027-01-18

Maths Ch 1 Coordinates · AI ICT Skills · Maths Ch 12 Quadrilaterals

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Q1 [3] **Question:** Three vertices of a rectangle are (1, 1), (6, 1) and (6, 4). Find the fourth vertex, the length, breadth and area of the rectangle.

Expected answer / marking points: Fourth vertex (1, 4) [1]; length 5, breadth 3 [1]; area 15 sq units [1].

Q2 [2] **Question:** Differentiate between hardware and software with one example each.

Expected answer / marking points: Hardware: physical parts (keyboard, CPU) [1]; software: programs/instructions (MS Word, OS) [1].

Q3 [2] **Question:** Write two safe practices while using the internet.

Expected answer / marking points: Strong passwords, don't share personal details, avoid unknown links/downloads, log out on shared devices (any two) [2].

Q4 [3] **Question:** Prove that in a parallelogram, opposite sides are equal.

Expected answer / marking points: Draw diagonal AC [0.5]; in $\triangle ABC$ and $\triangle CDA$: $\angle BAC = \angle DCA$, $\angle BCA = \angle DAC$ (alternate angles), AC common [1.5]; ASA congruence $\Rightarrow$ AB = CD, BC = DA [1].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Tuesday, 19 January 2027 · 10 marks

Test ID: D2027-01-19

Hindi पाठ 1 · SST Ch 1 · Science Ch 1 Exploration

Q1 [1]

Question: हीरा और मोती का मालिक कौन था?

Expected answer / marking points: झूरी।

Q2 [2]

Question: कहानी के आधार पर हीरा और मोती की गहरी मित्रता के दो उदाहरण लिखिए।

Expected answer / marking points: मूक-भाषा में बातचीत; एक-दूसरे को चाटकर प्रेम जताना; संकट में साथ न छोड़ना (कोई दो) [1 + 1]।

Q3 [1]

Question: 'दो बैलों की कथा' के लेखक कौन हैं?

Expected answer / marking points: प्रेमचंद।

Q4 [1]

Question: Name the four disciplines of Social Science in your Grade 9 textbook.

Expected answer / marking points: History, Geography, Political Science, Economics.

Q5 [1]

Question: Give one reason why studying Social Science helps a citizen.

Expected answer / marking points: Understand how government/laws work; participate responsibly; ask informed questions; respect diversity (any one).

Q6 [1]

Question: Name the four disciplines of Social Science studied in Grade 9.

Expected answer / marking points: History, Geography, Political Science, Economics.

Q7 [1]

Question: Which discipline studies how people make choices about scarce resources?

Expected answer / marking points: Economics.

Q8 [2]

Question: Write the steps of the scientific method in order.

Expected answer / marking points: Observation → question → hypothesis → experiment → analysis of results → conclusion (and share/repeat) [2].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Wednesday, 20 January 2027 · 10 marks

Test ID: D2027-01-20

SST Ch 2 Earth's surface · Science Ch 2 Cell · Maths Ch 2 Linear Polynomials

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Q1 [1]

Question: Name the three main layers of the Earth.

Expected answer / marking points: Crust, mantle, core.

Q2 [2]

Question: Distinguish between internal (endogenic) and external (exogenic) forces with one example of each.

Expected answer / marking points: Internal forces act from inside and build/move the crust — earthquakes, volcanoes, folding [1]; external forces wear the surface down — weathering, erosion by rivers, wind, glaciers [1].

Q3 [1]

Question: Which agent of gradation forms moraines?

Expected answer / marking points: Glaciers.

Q4 [1]

Question: Which organelle is the site of cellular respiration?

Expected answer / marking points: Mitochondria.

Q5 [2]

Question: Name two structures present in a plant cell but absent in an animal cell. Write the function of any one.

Expected answer / marking points: Cell wall, chloroplasts, large central vacuole (any two) [1]; function, e.g., chloroplast — photosynthesis; cell wall — rigidity [1].

Q6 [1]

Question: Write the slope and y-intercept of the line $y = 5x - 2$.

Expected answer / marking points: Slope 5, y-intercept −2.

Q7 [2]

Question: A linear polynomial p(x) has slope −3 and y-intercept 4. Write p(x) and find p(2).

Expected answer / marking points: $p(x) = -3x + 4$ [1]; $p(2) = -6 + 4 = -2$ [1].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Thursday, 21 January 2027 · 10 marks

Test ID: D2027-01-21

Maths Ch 3 World of Numbers · AI Communication Skills · English Unit 1

Q1 [3] **Question:** Simplify $(\sqrt{5} + \sqrt{2})^2$ and $(\sqrt{5} + \sqrt{2})(\sqrt{5} - \sqrt{2})$. Is each result rational or irrational?

Expected answer / marking points: $(\sqrt{5} + \sqrt{2})^2 = 7 + 2\sqrt{10}$ — irrational [1.5]; $(\sqrt{5} + \sqrt{2})(\sqrt{5} - \sqrt{2}) = 5 - 2 = 3$ — rational [1.5].

Q2 [1] **Question:** Write the stages of the communication cycle in the correct order.

Expected answer / marking points: Sender → Message → Channel → Receiver → Feedback.

Q3 [1] **Question:** Give one example of non-verbal communication.

Expected answer / marking points: Any: smile, nodding, gesture, eye contact, posture.

Q4 [2] **Question:** Write any two of the 7 Cs of communication with their meaning.

Expected answer / marking points: e.g., Clear — easy to understand; Concise — short, to the point; Correct; Courteous — polite; Complete; Coherent; Concrete (any two with meaning) [1 + 1].

Q5 [1] **Question:** Which story did the grandmother want to read herself in "How I Taught My Grandmother to Read"?

Expected answer / marking points: *Kashi Yatre* (by Triveni), in a weekly magazine.

Q6 [2] **Question:** Why was the grandmother in tears when the narrator came back from the wedding?

Expected answer / marking points: She could not read the next part of *Kashi Yatre* herself and felt helpless and dependent [2].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Friday, 22 January 2027 · 10 marks

Test ID: D2027-01-22

Hindi पाठ 3 संवादहीन · SST Ch 3 Atmosphere layers · Science Ch 3 Plant tissues

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Q1 [1]

Question: 'संवादहीन' कहानी के लेखक कौन हैं?

Expected answer / marking points: शेखर जोशी।

Q2 [1]

Question: कहानी के दो मुख्य पात्र कौन हैं?

Expected answer / marking points: ताई और तोता मिट्ठू।

Q3 [2]

Question: ताई के लिए मिट्ठू केवल एक तोता क्यों नहीं था?

Expected answer / marking points: अकेली ताई के लिए मिट्ठू बातचीत (संवाद) का माध्यम था [1]; वह उसकी ममता का केंद्र था [1]।

Q4 [1]

Question: About what percentage of the atmosphere is oxygen?

Expected answer / marking points: About 21%.

Q5 [2]

Question: Name the layer in which almost all weather phenomena occur. Write two features of this layer.

Expected answer / marking points: Troposphere [1]; ~12 km high, temperature falls with height, has most water vapour and clouds [1].

Q6 [1]

Question: Which plant tissue carries water from the roots to the leaves?

Expected answer / marking points: Xylem.

Q7 [2]

Question: Write two features of meristematic cells. Name any two types of meristem by location.

Expected answer / marking points: Thin walls, dense cytoplasm, large nucleus, divide repeatedly (any two) [1]; apical, lateral, intercalary (any two) [1].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Saturday, 23 January 2027 · 10 marks

Test ID: D2027-01-23

SST Ch 4 Early humans · Science Ch 4 Motion · Maths Ch 4 Identities

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Q1 [1]

Question: In which state are the Bhimbetka rock shelters located?

Expected answer / marking points: Madhya Pradesh.

Q2 [2]

Question: Write two important changes in human life during the Neolithic period.

Expected answer / marking points: Farming and domestication; permanent villages; pottery; weaving; polished tools (any two) [1 + 1].

Q3 [3]

Question: A train moving at 25 m s^{-1} applies brakes and stops in 10 s with uniform retardation. Find the retardation and the distance travelled before stopping.

Expected answer / marking points: $a = (0 - 25)/10 = -2.5$ m s^{-2} [1.5]; $s = ut + \frac{1}{2}at^2 = 250 - 125 = 125$ m [1.5].

Q4 [2]

Question: Expand $(a + 2b - c)^2$.

Expected answer / marking points: $a^2 + 4b^2 + c^2 + 4ab - 4bc - 2ca$ [2].

Q5 [2]

Question: If $a + b = 7$ and $ab = 10$, find $a^2 + b^2$.

Expected answer / marking points: $a^2 + b^2 = (a + b)^2 - 2ab$ [1] $= 49 - 20 = 29$ [1].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Maths · Full Paper 1 (Annual pattern) · Answer key

Sunday, 24 January 2027 · 80 marks

Test ID: T2027-01-24

Class 9 · Ganita Manjari Part I + II — poora syllabus (Ch 1–14)

Q1 **[1]**

Question: The point (0, –7) lies:

(a) on the x-axis
(b) on the y-axis
(c) in quadrant III
(d) in quadrant IV

Expected answer / marking points: (b) on the y-axis

Q2 **[1]**

Question: The x-coordinate (abscissa) of the point (–4, 9) is:

(a) 9
(b) –4
(c) 4
(d) –9

Expected answer / marking points: (b) –4

Q3 **[1]**

Question: Which of the following is a linear polynomial?

(a) $x^2 + 1$
(b) $3 - 2x$
(c) x^3
(d) $1/x$

Expected answer / marking points: (b) 3 – 2x

Q4 **[1]**

Question: For p(x) = 3x – 6, the value of x for which p(x) = 0 is:

(a) –2
(b) 3
(c) 2
(d) 6

Expected answer / marking points: (c) 2

Q5 **[1]**

Question: 0.333... written as p/q is:

(a) 3/10
(b) 1/3
(c) 33/100
(d) 3/100

Expected answer / marking points: (b) 1/3 — x = 0.333..., 10x = 3.333... ⇒ 9x = 3

Q6 **[1]**

Question: $\sqrt{12} \times \sqrt{3} =$

(a) 6
(b) $\sqrt{15}$
(c) 36
(d) $3\sqrt{2}$

Expected answer / marking points: (a) 6

Q7 **[1]**

Question: $(x + 2)(x + 3) =$

(a) $x^2 + 6$
(b) $x^2 + 5x + 6$
(c) $x^2 + 6x + 5$
(d) $x^2 + 5x + 5$

Expected answer / marking points: (b) $x^2 + 5x + 6$

Q8 [1] **Question:** Using an identity, 105 × 95 =

(a) 9975 (b) 10025

(c) 9925 (d) 9995

Expected answer / marking points: (a) 9975 — $100^2 - 5^2$

Q9 [1] **Question:** An arc subtends 100° at the centre of a circle. The angle it subtends at a point on the remaining part of the circle is:

(a) 100°

(b) 200°

(c) 50°

(d) 80°

Expected answer / marking points: (c) 50°

Q10 [1] **Question:** The circumference of a circle of diameter 14 cm is:

(a) 22 cm (b) 44 cm

(c) 88 cm (d) 154 cm

Expected answer / marking points: (b) 44 cm

Q11 [1] **Question:** Two coins are tossed. The probability of getting no head is:

(a) 1/2 (b) 1/4

(c) 3/4 (d) 0

Expected answer / marking points: (b) 1/4

Q12 [1] **Question:** The n-th term of the GP 3, 9, 27, ... is:

(a) 3n (b) 3^n

(c) n^3 (d) 3^{n-1}

Expected answer / marking points: (b) 3^n

Q13 [1] **Question:** 1 + 2 + 3 + ... + 20 =

(a) 200 (b) 210

(c) 220 (d) 190

Expected answer / marking points: (b) 210

Q14 [1] **Question:** A counterexample to "All prime numbers are odd" is:

(a) 3 (b) 9

(c) 2 (d) 1

Expected answer / marking points: (c) 2

Q15 [1] **Question:** The weighted mean of 10, 20, 30 with weights 1, 2, 3 is about:

(a) 20 (b) 23.33

(c) 25 (d) 30

Expected answer / marking points: (b) 23.33 — (10 + 40 + 90) ÷ 6

Q16 [1] **Question:** Using Euclid's algorithm, gcd(48, 18) =

(a) 3 (b) 6

(c) 9 (d) 18

Expected answer / marking points: (b) 6

Q17 [1] **Question:** The diagonals of a rhombus:

(a) are always equal (b) bisect each other at right angles

(c) do not bisect each other (d) are parallel

Expected answer / marking points: (b) bisect each other at right angles

Q18 [1] **Question:** Which line is parallel to $y = 3x + 1$?

(a) $y = -3x + 1$ (b) $y = x + 3$

(c) $y = 3x - 5$ (d) $y = (1/3)x$

Expected answer / marking points: (c) $y = 3x - 5$

Q19 [1] **Question: A:** Every rational number is an integer.
R: Every integer n can be written as n/1.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (d) — 1/2 is rational but not an integer.

Q20 [1] **Question: A:** The pair $2x + 3y = 5$ and $4x + 6y = 7$ has no solution.
R: Two lines are parallel when $a_1/a_2 = b_1/b_2 \neq c_1/c_2$.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (a)

Q21 [2] **Question:** Find k if $x = 2$, $y = -1$ is a solution of $3x + ky = 8$.

Expected answer / marking points: $6 - k = 8$ [1] $\Rightarrow k = -2$ [1].

Q22 [2] **Question:** Find the curved surface area of a cone of radius 7 cm and slant height 10 cm.

Expected answer / marking points: $\pi rl = 22/7 \times 7 \times 10$ [1] $= 220\ cm^2$ [1].

Q23 [2] **Question:** Find the 12th term of the AP $-5, -1, 3, 7, \ldots$

Expected answer / marking points: $a = -5$, $d = 4$ [1]; $t_{12} = -5 + 11 \times 4 = 39$ [1].

Q24 [2] **Question:** A die is thrown 60 times and a six comes up 12 times. Find the experimental probability of a six and compare it with the theoretical probability.

Expected answer / marking points: Experimental = 12/60 = 1/5 [1]; theoretical = 1/6; experimental is slightly more — they get closer with more trials [1].

Q25 [2] **Question:** Simplify $(x^2 - 4)/(x^2 + x - 6)$.

Expected answer / marking points: $(x - 2)(x + 2) / ((x + 3)(x - 2))$ [1] = $(x + 2)/(x + 3)$, $x \neq 2$ [1].

Q26 [3] **Question:** Express 0.2353535... (35 repeating) in the form p/q.

Expected answer / marking points: x = 0.23535...; 10x = 2.3535..., 1000x = 235.3535... [1]; 990x = 233 [1] ⇒ x = 233/990 [1].

Q27 [3] **Question:** In quadrilateral ABCD, AB = 3 cm, BC = 4 cm, CD = 4 cm, DA = 5 cm and diagonal AC = 5 cm. Find its area.

Expected answer / marking points: △ABC: 3, 4, 5 is right-angled ⇒ area 6 cm^2 [1]. △ACD: s = 7, area = $\sqrt{(7 \times 2 \times 3 \times 2)} = \sqrt{84} \approx 9.17$ cm^2 [1.5]. Total ≈ 15.17 cm^2 [0.5].

Q28 [3] **Question:** A chord of length 24 cm is at a distance of 5 cm from the centre of a circle. Find the radius. Also find the length of a chord at a distance of 12 cm from the centre.

Expected answer / marking points: $r = \sqrt{(12^2 + 5^2)} = 13$ cm [1.5]; half-chord = $\sqrt{(169 - 144)} = 5$ ⇒ chord = 10 cm [1.5].

Q29 [3] **Question:** A bag has 2 white and 3 black balls. A ball is drawn, put back, and a second ball is drawn. Find P(both black) and P(balls of different colours).

Expected answer / marking points: P(B) = 3/5, P(W) = 2/5 [0.5]; P(BB) = 9/25 [1]; P(different) = 2 × 2/5 × 3/5 = 12/25 [1.5].

Q30 [3] **Question:** In parallelogram ABCD, the bisectors of ∠A and ∠B meet at P. Prove that ∠APB = 90°.

Expected answer / marking points: ∠A + ∠B = 180° (co-interior angles, AD ∥ BC) [1]; ½∠A + ½∠B = 90° [1]; in △APB, ∠APB = 180° − 90° = 90° [1].

Q31 [3] **Question:** A road roller has diameter 84 cm and length 120 cm. It takes 500 complete revolutions to level a playground. Find the area of the playground in m^2.

Expected answer / marking points: Area per revolution = $2\pi rh = 2 \times 22/7 \times 42 \times 120 = 31{,}680$ cm^2 [1.5]; × 500 = 1,58,40,000 cm^2 [1] = 1584 m^2 [0.5].

Q32 [5] **Question:** Draw the graphs of $2x + y = 8$ and $x - y = 1$. (i) Find their point of intersection. (ii) Find the vertices and area of the triangle formed by these lines and the x-axis.

ATHVA / OR

If 2 is added to both the numerator and denominator of a fraction, it becomes 9/11. If 3 is added to both, it becomes 5/6. Find the fraction.

Expected answer / marking points: Tables + graph [2]. (i) $3x = 9 \Rightarrow (3, 2)$ [1]. (ii) Vertices (3, 2), (4, 0), (1, 0) [1]; area = ½ × 3 × 2 = 3 sq units [1].
OR: Let fraction x/y: $11(x + 2) = 9(y + 2) \Rightarrow 11x - 9y = -4$ [1]; $6(x + 3) = 5(y + 3) \Rightarrow 6x - 5y = -3$ [1]; solving: $x = 7$, $y = 9$ [2]; fraction 7/9; check 9/11 ✓, 10/12 = 5/6 ✓ [1].

Q33 [5] **Question:** (i) If $x - y = 3$ and $xy = 10$, find $x^3 - y^3$. (ii) Factorise $27a^3 + 64b^3$. (iii) Evaluate 103^3 using an identity.

Expected answer / marking points: (i) $x^3 - y^3 = (x - y)^3 + 3xy(x - y) = 27 + 90 = 117$ [2]. (ii) $(3a + 4b)(9a^2 - 12ab + 16b^2)$ [1.5]. (iii) $(100 + 3)^3 = 1000000 + 90000 + 2700 + 27 = 1092727$ [1.5].

Q34 [5] **Question:** Prove that the angle subtended by an arc at the centre of a circle is double the angle subtended by it at any point on the remaining part of the circle.

ATHVA / OR

Prove that equal chords of a circle subtend equal angles at the centre.

Expected answer / marking points: Figure: arc AB, centre O, point P on remaining part; join PO and extend to Q [1]. In $\triangle OAP$, $OA = OP \Rightarrow \angle OAP = \angle OPA$; exterior $\angle AOQ = 2\angle OPA$ [1.5]. Similarly $\angle BOQ = 2\angle OPB$ [1.5]. Adding: $\angle AOB = 2\angle APB$ [1].
OR: Figure: chords AB = CD, centre O [1]. In $\triangle AOB$ and $\triangle COD$: OA = OC, OB = OD (radii), AB = CD (given) [2]; SSS $\Rightarrow \triangle AOB \cong \triangle COD$ [1]; $\Rightarrow \angle AOB = \angle COD$ (CPCT) [1].

Q35 [5] **Question:** A toy is a hemisphere of radius 3.5 cm with a cone of the same radius on top. The total height of the toy is 15.5 cm. Find (i) the slant height of the cone (ii) the total surface area of the toy (iii) its volume.

Expected answer / marking points: Cone height = 15.5 − 3.5 = 12 cm [0.5]; (i) $l = \sqrt{(144 + 12.25)} = 12.5$ cm [1]. (ii) $\pi rl + 2\pi r^2 = 137.5 + 77 = 214.5$ cm^2 [1.5]. (iii) $\frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3 = 154 + 89.83 \approx 243.83$ cm^3 [2].

Q36 [4] **Question:**

> A school auditorium has 20 seats in the first row, 22 in the second, 24 in the third, and so on.

(i) How many seats are there in the 10th row? [1]
(ii) Which row has 60 seats? [1]
(iii) By pairing the first and the last of the first 10 rows (row 1 + row 10, row 2 + row 9, ...), find the total number of seats in the first 10 rows. [2]

Expected answer / marking points: (i) $20 + 9 \times 2 = 38$ [1]. (ii) $20 + 2(n - 1) = 60 \Rightarrow n = 21$ [1]. (iii) Each pair = 20 + 38 = 58, 5 pairs [1] $\Rightarrow$ 290 seats [1].

Q37 **Question:**

[4]

> Two branches of a stationery shop sold pens in January and February: Branch A sold 120 and 80; Branch B sold 60 and 140.

(i) What were the total pen sales in January? [1]
(ii) In a 100% stacked bar chart for January, what percentage of the bar would Branch A take? [1]
(iii) Branch A sold its 200 pens at ₹10 each and Branch B its 200 pens at ₹12 each. Find the average selling price of all 400 pens. If Branch B had sold 300 pens at ₹12, what would the average be? [2]

Expected answer / marking points: (i) 120 + 60 = 180 [1]. (ii) 120/180 × 100 ≈ 66.7% [1]. (iii) (2000 + 2400)/400 = ₹11 [1]; (2000 + 3600)/500 = ₹11.20 [1].

Q38 **Question:**

[4]

> Riya finds all divisors of 36 with this algorithm: for each j from 1 to 36, if j divides 36, add j to the list.

(i) Write the list she gets. [1]
(ii) How many numbers does she check? [1]
(iii) Her brother checks only j = 1 to 6 and, whenever j divides 36, writes both j and 36/j. How many checks does he make, and why does his method still find every divisor? [2]

Expected answer / marking points: (i) 1, 2, 3, 4, 6, 9, 12, 18, 36 [1]. (ii) 36 [1]. (iii) 6 checks [1]; every divisor d bigger than 6 pairs with 36/d, which is smaller than 6 and is checked, so d gets written too [1].

Score: ___ / 80 Checked by: _____

Weak topic aur correction:

Parent Sunday handover checklist

- [ ] Agle Monday–Sunday ka student pack print karke diya.
- [ ] Parent solved key alag rakhi; student copy ke saath share nahi ki.
- [ ] Pichhle week ka pending kaam aur 3 weak topics review kiye.
- [ ] School exam date sheet aur tuition timings se schedule match kiya.

Agli priority / teacher se poochhna: